MILITARY SPECIFICATION SHEET

SWITCHES, SENSITIVE, LIMIT, PLUNGER, 7 AMPERES OR LOW LEVEL, RESILIENT SEAL, FLUID RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.

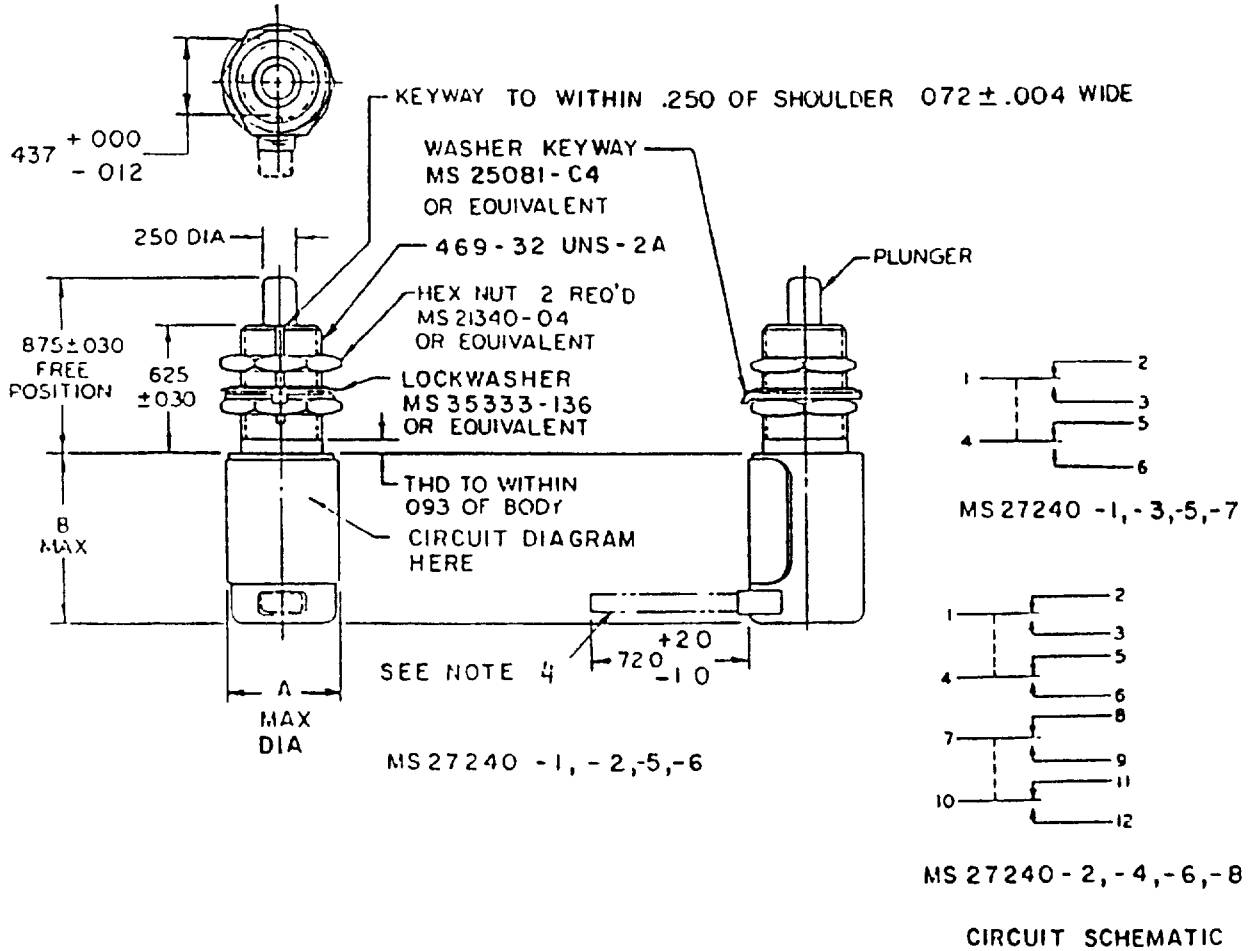
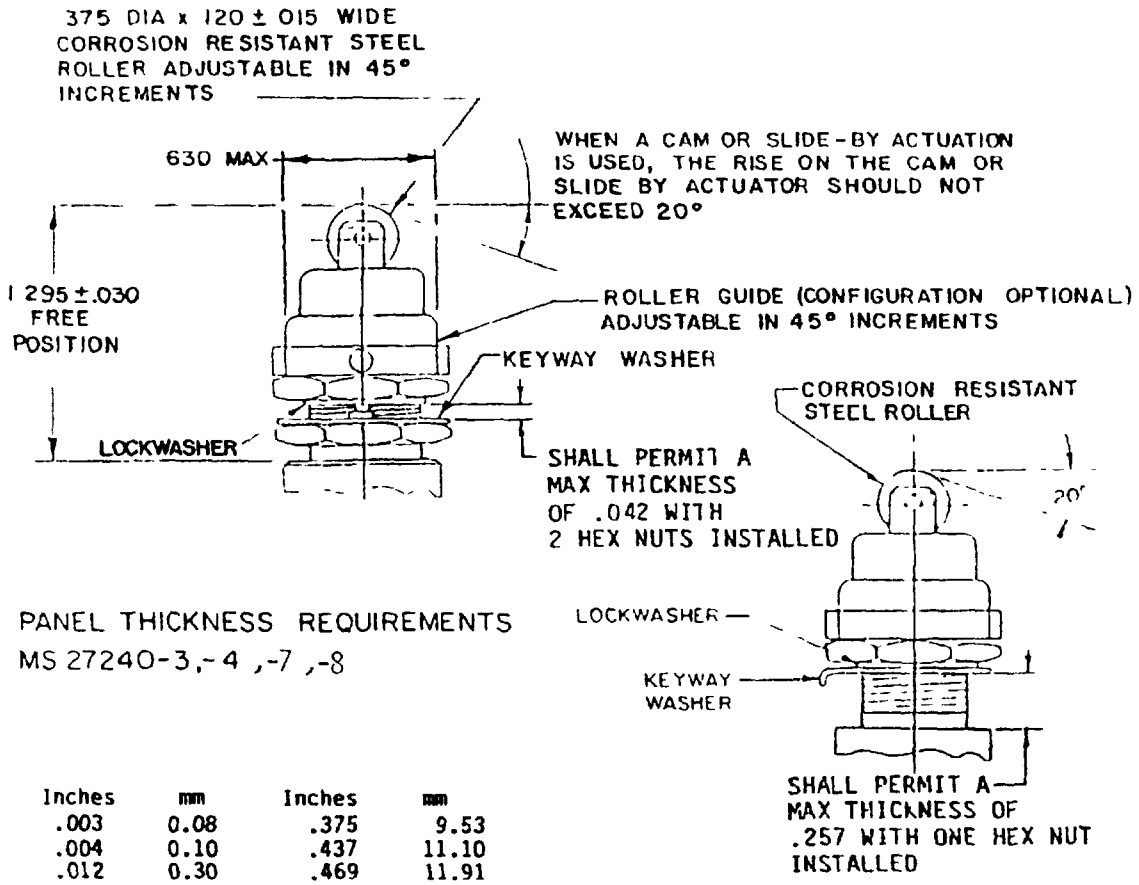


FIGURE 1 Configuration and dimensions

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PANEL THICKNESS REQUIREMENTS
MS 27240-3,-4 ,-7 ,-8

Inches	mm	Inches	mm
.003	0.08	.375	9.53
.004	0.10	.437	11.10
.012	0.30	.469	11.91
.015	0.38	.625	15.88
.030	0.76	.630	16.00
.031	0.79	.680	17.27
.042	1.06	.876	22.23
.072	1.83	1.0	25.
.093	2.36	1.295	32.89
.120	3.05	2.0	50.
.250	6.35	72.0	1828.
.257	6.53		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
3. Exact shape of switch optional provided dimensions specified are not exceeded.
4. Leads shall be M22759/7-20-9 wire marked in accordance with MIL-W-5088 with the switch circuit identification number followed by wire gauge number (1-20, 2-20, etc.). Leads may be stripped .500 maximum for the purpose of inspection of electrical parameters Leads shall be potted to provide stress relief.
5. Metric equivalents are given for general information only

FIGURE 1. Configuration and dimensions - Continued.

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REQUIREMENTS

Dimensions and configuration: See figure 1 and table I.

TABLE I. Part or Identifying Number (PIN) and characteristics.

PIN	A diameter maximum	B diameter maximum	Actuator	Lead wires, number required	Contact material
MS27240-1	.720 (18.29)	1.000 (25.40)	Plunger	6	Silver
MS27240-2	1.031 (26.19)	1.200 (30.48)	Plunger	12	"
MS27240-3	.720 (18.29)	1.000 (25.40)	Roller plunger	6	"
MS27240-4	1.031 (26.19)	1.200 (30.48)	Roller plunger	12	"
MS27240-5	.720 (18.29)	1.000 (25.40)	Plunger	6	Gold
MS27240-6	1.031 (26.19)	1.200 (30.48)	Plunger	12	"
MS27240-7	.720 (18.29)	1.000 (25.40)	Roller plunger	6	"
MS27240-8	1.031 (26.19)	1.200 (30.48)	Roller plunger	12	"

Enclosure design: Four (resilient) All entrances to the switch cavity except through the actuator bushing shall be sealed by fusion of glass-to-metal, metal-to-metal, or ceramic-to-metal and the lead wires shall be potted to provide stress relief.

Temperature characteristic: One (-55°C to +85°C).

Shock type. M (100 G).

Sinusoidal vibration grade. Two (10 to 2,000 Hz).

Material:

Plastic: Type CFG in accordance with MIL-M-14 may be used internally.

Finish: Switch housing shall be processed to resist corrosion.

Maximum weight with leads:

Two pole. .50 pound.

Four pole: .85 pound.

Operating characteristics:

Actuating force: 9 ±3 pounds.

Release force: 4 pounds minimum.

Movement differential: .020 inch (0.50 mm) maximum.

Pretravel: .040 inch (1.01 mm) maximum.

Overtravel: .125 inch (3.17 mm) minimum.

Full overtravel force: 30 pounds maximum

} ±20 percent variation from specified values acceptable after test.

Coincidence of operating and releasing points: All poles shall transfer within .010 inch of plunger travel.

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Strength of actuating means 30 pounds

Contact resistance

Silver contact switches Not applicable

Gold contact switches In accordance with MIL-S-8805, plus .015 ohm maximum per lead per foot.

Dielectric withstanding voltage.

At atmospheric pressure 1,000 V rms.

At reduced barometric pressure: 50,000 feet, 400 V rms.

Mechanical endurance. 25,000 cycles.

Electrical endurance. 25,000 cycles

Electrical ratings: See tables II and III.

Low level circuit: Applicable for gold contact switches only, MS27240-5 through -8. 25,000 cycles.

Logic level circuit: Applicable for gold contact switches only, MS27240-5 through MS27240-8: 25,000 cycles.

Fluid resistance: Except for the cut end of the lead wire, switches shall be submerged in each of the following fluids for 2 minutes to 2 minutes 30 seconds, which shall consist of one cycle (one cycle is 10 minutes to 12 minutes 30 seconds total). Each switch shall be subjected to three cycles.

a. Turbine fuel (MIL-T-5624).

b. Hydraulic fluid (SAE AS1241A).

c. Coolanol 1/ (MIL-C-47220).

d. Ethylene glycol (MIL-E-9500).

e. Lubricating oil (MIL-L-7808).

After each immersion, the excess fluid shall be blown off the external surfaces of the switch with an air jet. Following the third cycle, the switch shall be subjected to and shall meet the requirements for terminal strength, dielectric withstanding voltage, insulation resistance, operating characteristics, seal tests, and marking visibility.

Marking: The circuit schematic shall be marked on the switch case.

PIN: See table I.

1/ Monsanto Company registered trademark.

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TABLE II. Electrical ratings, silver contacts.

Load	Sea level 28 V dc	50,000 feet 28 V dc
	(amperes)	(amperes)
Resistive	7	7
Inductive	4	2.5
Motor	4	4 1/

1/ Application information only.

TABLE III. Electrical ratings, gold contacts.

Load	Sea level			50,000 feet	
	Power circuit 28 V dc	Electronic logic 5 V dc	Low level 30 millivolts dc or peak ac	Power circuit 28 V dc	Electronic logic 5 V dc
	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)
Resistive	1.0	0.01	0.01	1.0	0.01
Inductive	0.5	---	---	0.5	---

Qualification inspection:

Group submission: See table IV.

Group A inspection:

Seal test: Only watertight test shall be performed.

TABLE IV. Qualification inspection (group submission).

Examination or test	Samples	Extent of approval
Qualification inspection table of MIL-S-8805	MS27240-4 (24 units)	All
Groups I, VII, X, and XI of qualification inspection table of MIL-S-8805 1/	MS27240-8 (8 units)	
Visual and mechanical examination Operating characteristics	MS27240-1, -2, -3 (2 units each)	

1/ Logic level circuit: Test sea level ratings only.
Electrical endurance (power circuit): Test altitude ratings only.

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Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians.
Army - EK
Navy - EC
Air Force - 85

Review activities
Army - AR
Navy - AS
Air Force - 99
DLA - ES

User activities.
Army - AV, MI
Navy - MC
Air Force - 19

Preparing activity.
Navy - EC

Agent
DLA - ES

(Project 5930-1413-02)

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